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Review

Machine Learning Applications in Building Energy Systems: Review and Prospects

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Abstract: Building energy systems (BESs) are essential for modern infrastructure but face significant challenges in equipment diagnosis, energy consumption prediction, and operational control. The complexity of BESs, coupled with the increasing integration of renewable energy sources, presents difficulties in fault detection, accurate energy forecasting, and dynamic system optimisation. Traditional control strategies struggle with low efficiency, slow response times, and limited adaptability, making it difficult to ensure reliable operation and optimal energy management. To address these issues, researchers have increasingly turned to machine learning (ML) techniques, which offer promising solutions for improving fault diagnosis, energy scheduling, and real-time control in BESs. This review provides a comprehensive analysis of ML techniques applied to fault diagnosis, energy consumption prediction, energy scheduling, and operational control. According to the results of analysis and literature review, supervised learning methods, such as support vector machines and random forest, demonstrate high classification accuracy for fault detection but require extensive labelled datasets. Unsupervised learning approaches, including principal component analysis and clustering algorithms, offer robust fault identification capabilities without labelled data but may struggle with complex nonlinear patterns. Deep learning techniques, particularly convolutional neural networks and long short-term memory models, exhibit superior accuracy in energy consumption forecasting and real-time system optimisation. Reinforcement learning further enhances energy management by dynamically adjusting system parameters to maximise efficiency and cost savings. Despite these advancements, challenges remain in terms of data availability, computational costs, and model interpretability. Future research should focus on improving hybrid ML models, integrating explainable AI techniques, and enhancing real-time adaptability to evolving energy demands. This review also highlights the transformative potential of ML in BESs and outlines future directions for sustainable and intelligent building energy management.

Keywords: machine learning; building energy systems; fault diagnosis; energy consumption prediction; operational control

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1. Introduction

In recent years, global warming has become an increasingly prominent issue, drawing extensive attention from the international community toward reducing greenhouse gas emissions, particularly carbon dioxide (CO_2). According to the 2023 report by the International Energy Agency on global energy use and carbon emissions in the building sector in

2022, buildings account for approximately 40% of the world's total energy consumption, making it the second-largest energy-consuming sector globally [1]. The operational phase alone consumes 27% of global end-use energy and contributes 17% of CO_2 emissions, positioning it as a critical area for energy conservation and emission reduction [1]. Building energy systems (BESs) integrate renewable energy systems and comprehensive energy systems with buildings, reshaping the supply–demand relationship. This enables buildings to transition from being pure energy consumers to becoming energy producers and distributors, paving the way for zero-carbon and net-zero energy buildings [2–4].

Researchers have conducted extensive studies on building energy systems, covering fault diagnosis, energy consumption prediction, and planning and operational control of building energy systems [5–7]. However, building energy systems still face numerous issues and challenges, which are specifically manifested in the following scenarios:

(1) Equipment Diagnosis: Building energy systems comprise a wide variety and large number of devices. Ensuring the reliable operation of these devices over long periods is fundamental to the safe operation of the entire system [5]. However, current fault diagnosis methods are inefficient and have high false-positive rates, making the rapid and accurate diagnosis of building energy equipment a significant challenge.
(2) Energy Consumption Prediction: As building energy loads continue to grow, the volatility and uncertainty of these loads have significantly increased, demanding higher accuracy in forecasting various energy loads. However, the current precision and robustness of energy consumption prediction in building energy systems remain low [8].
(3) Operational Control: With the increasing concentration of building energy devices, the complexity of energy system operation and control has become more prominent. It is challenging to dynamically adjust system parameters and promptly detect potential faults. Therefore, the level of system intelligence needs to be further enhanced [9].

The existing literature has proposed control strategies and application methods based on optimised operational algorithms for building energy systems in areas such as equipment diagnosis, energy consumption prediction, and planning operations [10]. However, traditional methods have the following issues:

(1) Traditional control strategies fail to effectively integrate renewable energy, resulting in low utilisation rates and an inability to leverage the potential advantages of renewable energy within the system. This creates limitations in energy dispatching [11].
(2) During high-load periods or emergencies in building energy systems, traditional algorithms cannot effectively meet specific energy dispatch demands. This increases operational risks and makes it difficult to ensure the stability and reliability of energy dispatching [12].
(3) Control strategies for building energy systems exhibit slow response times, low energy utilisation efficiency, and poor diagnostic accuracy [13]. As building energy systems typically consist of multiple energy systems and renewable energy generation technologies, they are inherently nonlinear systems, which traditional methods struggle to handle effectively [14].

Machine learning (ML) aims to enable computer systems to learn and reason logically, extracting knowledge and experience from existing data to perform pattern recognition, prediction, and decision making [15]. ML technologies have already been widely applied in the field of building energy systems [16]. Compared to traditional methods, ML offers faster fault identification and higher system prediction accuracy [17]. Additionally, ML significantly enhances the accuracy and stability of energy consumption prediction through advancements in time series data analysis, feature extraction, data preprocessing, and load

forecasting modelling [18]. Therefore, the introduction of ML into the development of building energy systems can improve fault diagnosis efficiency and energy consumption prediction accuracy, facilitate the innovation and restructuring of existing energy systems, and enable dynamic monitoring of building energy systems. This reduces energy transaction risks, improves energy management systems, and ultimately enhances overall energy utilisation efficiency in buildings.

For instance, deep reinforcement learning has been successfully applied in energy scheduling, enabling cost-effective energy distribution and improved operational efficiency [19]. Additionally, physics-informed neural networks (PINNs) have been utilised to enhance the accuracy of thermal energy modelling, outperforming conventional machine learning techniques in complex BES scenarios [20]. Moreover, the integration of generative adversarial networks (GANs) with energy forecasting models has significantly improved predictive accuracy by generating high-resolution energy consumption patterns [21]. The role of ML has also expanded to real-time monitoring and digital twin technologies, where generative AI and IoT-enabled smart sensors facilitate adaptive control strategies and predictive maintenance [22,23]. Furthermore, edge computing and federated learning approaches have emerged as promising techniques for decentralised BES optimisation, reducing data transmission costs while maintaining high computational efficiency [24]. These technological advancements illustrate the transformative impact of ML in enhancing BES efficiency; optimising control strategies; and driving the evolution of intelligent, autonomous building energy systems.

This paper reviews the current state of research on the application of machine learning (ML) technologies in various areas, including fault diagnosis of building equipment, energy consumption prediction, optimisation of energy scheduling and operational control, and strategies for energy trading in building energy systems. It introduces relevant research advancements and key technical methods. The reviewing method is systematic, comprehensive, and practical, offering a clear overview of the current state of research and future directions in the application of machine learning to building energy systems. The literature was selected based on its relevance to the application of machine learning in building energy systems, with a focus on recent advancements, practical implementations, and significant contributions to fault diagnosis, energy consumption prediction, and operational control. Finally, the paper summarises the application methods of ML in building energy systems and outlines future research directions.

2. Machine Learning Applications in Fault Diagnosis for Building Energy Systems

Machine-learning-based fault detection and diagnosis (FDD) technologies [25] have demonstrated tremendous potential in improving the accuracy and response speed of fault diagnosis. By analysing large datasets, ML models can automatically extract critical features, identify complex fault patterns, and enable early warnings and precise classifications.

In recent years, various ML methods, including supervised learning, unsupervised learning, hybrid learning, and deep learning, have been widely applied to different fault diagnosis scenarios. Supervised learning (SL) is a machine learning approach where the model is trained using labelled data, meaning each input has a corresponding correct output, allowing the algorithm to learn the mapping between inputs and outputs [26]. It is commonly used for classification and regression tasks, such as predicting building energy consumption or detecting heating, ventilation, and air conditioning (HVAC) system failures [26]. Unsupervised learning (USL), on the other hand, is used when the data lack labels, requiring the model to identify patterns, clusters, or anomalies without predefined outcomes [27]. This method is useful for applications like energy consumption pattern

recognition and fault detection in energy systems [27]. Hybrid learning (HL) combines elements of both supervised and unsupervised learning, leveraging labelled data when available while also discovering hidden structures in unlabelled data [28]. It is particularly effective in complex scenarios, such as fault diagnosis in energy systems, where some data are labelled and other parts require clustering techniques for deeper insights [28]. Deep learning (DL) is a subset of machine learning that uses multi-layered neural networks to model complex patterns and nonlinear relationships in data, excelling in time series analysis, anomaly detection, and optimisation tasks for building energy management [29]. DL methods, such as convolutional neural networks (CNNs) and long short-term memory (LSTM), are widely applied in energy load forecasting and intelligent control of HVAC systems [30]. Sections 2.1 and 2.2 summarise the research progress on the application of ML in system-level and equipment-level fault diagnosis in building energy systems, respectively.

2.1. System-Level Fault Diagnosis

In modern building energy management, the heating, ventilation, and air conditioning (HVAC) system, as a core subsystem, directly affects overall energy consumption and environmental comfort through its operational efficiency and reliability. Fault detection and diagnosis (FDD) technology, by monitoring the operational status of the HVAC system in real time, can effectively identify potential faults and take timely maintenance measures, thereby reducing operating costs.

2.1.1. Supervised Learning

Supervised learning, one of the most commonly used techniques in the field of fault detection and diagnosis (FDD), achieves accurate fault type prediction and classification by utilising labelled datasets for model training. Es-sakali, Zoubir, Kaitouni, Mghazli, Cherkaoui, and Pfafferott [6] developed a predictive maintenance strategy based on machine learning, which optimised data quality through noise filtering models such as Kalman filtering, moving average, and Savitzky–Golay smoothing and selected ten key features for model training. The experimental results showed that the accuracy of the random forest (RF) model was 98%. Ebrahimifakhar et al. [31] proposed a data-driven FDD strategy based on statistical machine learning classification methods, modelling the FDD problem as a multi-class task to simultaneously detect and diagnose normal operating conditions and seven common faults. The experimental results indicated that the accuracy of the support vector machine (SVM) reached 96.2%. Chakraborty and Elzarka [32] proposed a real-time fault detection model based on the extreme gradient boosting (XGBoost) algorithm and dynamic threshold methods, using XGBoost to generate a predictive energy consumption model and dynamically adjusting the threshold via Chebyshev's inequality. The experimental results showed that this method improved the F1 score by 8.82% and 117.65%, respectively, compared to traditional fixed-threshold and existing dynamic-residual threshold methods across various datasets.

In practical fault diagnosis applications, while supervised learning methods can perform well when sufficient high-quality labelled data are available, they rely on large amounts of accurately labelled training data. This requirement makes supervised learning vulnerable when facing scenarios with insufficient or incomplete labelled data. Unsupervised learning, through self-learning and data-driven approaches, offers an effective alternative.

2.1.2. Unsupervised Learning

Unlike supervised learning, unsupervised learning does not rely on labelled data. Its core lies in discovering potential fault behaviours by identifying patterns in the data. Soultanzadeh et al. [33] proposed an unsupervised automated fault detection and diagnosis (AFDD) method based on principal component analysis (PCA) for the HVAC system in

light commercial buildings. By analysing unlabelled, noisy, and incomplete data and using reconstruction errors from historical data for fault detection and diagnosis, the experimental results showed that the method could effectively detect and isolate faults, providing information on fault location and severity, with an accuracy of 86%.

Tra et al. [34] proposed an unsupervised anomaly detection method based on a neural-network-based mixture of probabilistic principal component analysers (NN-MPPCA). By integrating neural networks into the mixture of probabilistic principal component analysers (MPPCA) framework and optimising model parameters using the backpropagation algorithm, the experimental results indicated that after data cleaning with NN-MPPCA, the accuracy of the fault diagnosis model increased to 99.5%, significantly improving by nearly 20% compared to directly using contaminated data (accuracy of 82.6%). Yan et al. [35] proposed an AFDD method based on a conditional Wasserstein generative adversarial network (CWGAN). By generating high-quality synthetic fault samples with a CWGAN to rebalance the training dataset, the experimental results showed that the classification accuracy for fault detection and fault diagnosis was 90% and 90.4%, respectively.

2.1.3. Hybrid Learning

In the development of FDD technology, hybrid learning methods have gradually become an important choice for addressing complex scenarios. Hybrid learning combines the advantages of supervised and unsupervised learning, making it particularly suitable for diagnosing tasks in systems with multiple coexisting faults and dynamic behaviour. Tun et al. [36] proposed a hybrid model based on RF and SVM (hybrid random forest–support vector machine, HRF–SVM), which uses RF to extract the most critical features and then inputs these features into SVM for binary classification. Experimental results show that this method significantly outperforms standalone RF or SVM models. Gharsellaoui et al. [37] introduced an FDD method based on multiscale interval principal component analysis (MSIPCA) and machine learning. The MSIPCA method is used to extract and reduce features from interval data, followed by SVM or K-nearest neighbour (KNN) classifiers for fault classification. Experiments show that this method achieves a classification accuracy of up to 97.53% in synthetic data and simulated heating systems. Albayati et al. [38] proposed a method that combines SVM and unsupervised learning, using k-NN and clustering methods to expand the training dataset to address the scarcity of labelled data. Experimental results indicate that this method achieves an average classification accuracy of 95.7% when handling multiple types of faults. These studies suggest that hybrid learning methods can effectively enhance diagnostic performance in complex scenarios. However, compared to traditional machine learning algorithms, hybrid learning models have higher training complexity and demand greater computational resources.

2.1.4. Deep Learning

In recent years, the emergence of deep learning has provided a new development direction for FDD technology. Compared with traditional machine learning, deep learning models have more complex and adaptive network architectures, as well as stronger feature extraction and function fitting capabilities. Taheri et al. [39] proposed a method combining deep recurrent neural networks (DRNNs) and hyperparameter optimisation. Through stacking models and optimising network structures with deep transitional output layers, the model performance was improved. Experimental results showed that the average accuracy on single-zone and multi-zone air handling unit datasets reached 91% and 87%, respectively, significantly outperforming RF and gradient boosting (GB) models. Tian et al. [40] proposed a fault diagnosis method for chillers based on automated machine learning and LSTM. By using an improved maximum relevance minimum redundancy algorithm for feature

selection and applying genetic algorithms to optimise the hyperparameters and network structure of LSTM, the method achieved a maximum diagnostic accuracy of 98.92% under various fault severities and complexities. Compared with traditional methods (e.g., SVMs, RNNs, BPNNs), the diagnostic accuracy for low-severity faults was improved by 25.92%. Li et al. [41] proposed a model based on a one-dimensional convolutional neural network (1D-CNN). By eliminating pooling layers, using soft sign activation functions, and fixing convolution kernel sizes to retain the sequence and features of the data, combined with an improved absolute gradient-weighted class activation mapping method, the model provides visual fault diagnosis standards and enhances interpretability. On the ASHRAE RP-1043 dataset, the diagnostic accuracy for typical chiller faults exceeded 98.5%.

Deep learning exhibits outstanding advantages in fault diagnosis of building energy systems. Its high prediction accuracy and strong time series processing capabilities enable it to effectively capture the nonlinear features and dynamic behaviours of complex systems. However, it relies on large-scale labelled data and high-performance computing resources.

The various machine learning methods for system-level fault diagnosis in building energy systems presented in this section, including supervised learning, unsupervised learning, hybrid learning, and deep learning, are summarised in Table 1.

Table 1. Comparison of machine learning methods for system-level fault diagnosis.

Types	Methods	Advantages	Limitations	References
Supervised Learning	RF	Handles small datasets; simple and easy to implement.	Complex data preprocessing; heavily relies on the quality of labeled data.	[6]
	SVM	High classification accuracy.	High computational complexity; requires sufficient labeled data support.	[31]
	XGBoost	Excellent real-time fault detection performance; significant F1 score improvement.	High demand for parameter tuning; time-consuming training process.	[32]
Unsupervised Learning	PCA	Suitable for unlabeled, noisy, and incomplete data.	Limited adaptability to nonlinear data.	[33]
	NN-MPPCA	High F1 scores and diagnostic accuracy with cleaned data.	High computational complexity; depends on the distribution of outliers.	[34]
	CWGAN	High classification accuracy for fault detection and multi-fault diagnostics.	Complex model training; depends on the quality of generated samples; requires significant computational resources.	[35]
Hybrid Learning	HRF–SVM	Significantly outperforms standalone RF or SVM models; combines the strengths of both methods.	Increased model complexity; high demands for feature extraction and classification parameter tuning.	[36]
	MSIPCA and SVM, KNN Hybrid	High classification accuracy; significantly reduces the computational complexity of high-dimensional data.	High preprocessing requirements for interval data; high demand for computational resources.	[37]
	SVM with k-NN and Clustering	High average classification accuracy; expands datasets to improve classification performance.	Increased model complexity; highly dependent on dataset expansion and clustering algorithms.	[38]

Table 1. *Cont.*

Types	Methods	Advantages	Limitations	References
Deep Learning	DRNN	Achieved average accuracies of 91% and 87% on single-zone and multi-zone air handling unit datasets, respectively; significantly outperforms RF and GB models.	High model complexity; involves extensive neural network architecture search and parameter tuning when optimising with GA.	[39]
	LSTM	Achieved average accuracies of 91% and 87% on single-zone and multi-zone air handling unit datasets, respectively; significantly outperforms RF and GB models.	Slow diagnostic speed.	[40]
	1D-CNN	High diagnostic accuracy; provides model transparency through visualisation analysis.	Requires extensive data support; limited adaptability to tasks beyond time series feature extraction.	[41]

2.2. Equipment-Level Fault Diagnosis

In equipment-level fault detection, supervised learning uses data from known faults and normal operating states to train models to determine whether the equipment is currently in a fault state. Hong et al. [42] proposed a novel decision framework based on a support vector machine (SVM) for power transformer fault diagnosis (PTFD) in building energy systems. This method achieves high diagnostic accuracy while reducing training time. Zhang et al. [43] introduced a new fault diagnosis classification optimisation method, TLSCA-SVM, which combines the sine–cosine algorithm, SVM, and transfer learning. This approach effectively operates under conditions of limited fault data to achieve the benefits of transfer learning. However, a review of current studies on SVM in equipment fault diagnosis reveals significant challenges in kernel function selection and optimisation, which limit its application. Biswal et al. [44] presented a fault diagnosis technique based on dual-tree complex wavelet transform (DT-CWT) and data mining methods. Kherif et al. [45] employed a K-nearest neighbour (KNN) classifier, integrating the decision tree principle to select the optimal input vector, thereby improving the diagnostic accuracy of power transformer faults in building energy systems. In the above methods, feature extraction and model training are conducted in isolation, making it difficult for these approaches to adapt to complex application scenarios.

In equipment-level fault detection, unsupervised learning can identify abnormal behaviours from operational data through methods like clustering analysis or anomaly detection, further determining whether the equipment is faulty. Wang et al. [46] proposed a fault detection method for building energy generators based on variational mode decomposition (VMD) and a stacked autoencoder (SAE). This approach effectively handles high-dimensional time series data, improving the accuracy and reliability of anomaly detection in building energy generators.

In the fault diagnosis domain, especially in the maintenance of building energy systems, insufficient training samples are a common issue. Lu et al. [47] introduced a method based on an improved conditional variational autoencoder (MCVAE) and a co-training (CT) framework to address the lack of training samples in chiller fault diagnosis for heating, ventilation, and air conditioning (HVAC) systems in commercial buildings. This method significantly improves the accuracy and efficiency of fault diagnosis. Clustering is one of

the most common unsupervised learning algorithms, which has recently gained attention from researchers in the field of fault diagnosis. Common clustering algorithms include K-means clustering, density-based clustering, and fuzzy clustering. Yang et al. [48] combined a genetic algorithm (GA) with density-based spatial clustering of applications with noise (DBSCAN) to propose a new method, GA-DBSCAN, for fault diagnosis of gas-insulated switchgear in building energy systems. This method greatly improves diagnostic accuracy for such equipment. Hou and Zhang [49] improved the density peak clustering algorithm by normalising the local density based on neighbouring data before identifying cluster centres, significantly enhancing the clustering results. Liu et al. [50] presented a fault diagnosis method combining a stacked autoencoder (SAE) and clustering algorithms for fault detection and classification in building energy rooftop photovoltaics. This method addresses the problem of rooftop photovoltaics being prone to various faults in harsh environments. Wen et al. [51] proposed an enhanced principal component analysis (PCA) method that combines DBSCAN and Savitzky–Golay filters to improve the accuracy and sensitivity of sensor fault detection in air conditioning systems.

In recent years, deep learning (DL) technology, combined with brain-inspired cognitive mechanisms for data processing, has overcome the bottleneck of traditional neural networks in practical applications. Compared to shallow machine learning methods such as random forests (RFs) and support vector machines (SVMs), fault diagnosis methods based on deep learning exhibit significantly stronger performance. Common deep learning methods include convolutional neural networks (CNNs), long short-term memory (LSTM), deep neural networks (DNNs), and deep belief networks (DBNs). Song et al. [52] proposed a CNN model based on convolution kernels, wide-convolution-kernel CNN (WKCNN). WKCNN exhibits robust noise resistance, maintaining excellent diagnostic performance when processing noisy vibration signals. As an important module of HVAC systems, faults in air handling units (AHUs) reduce indoor comfort, shorten equipment lifespan, and waste building energy. To address this, Zhang et al. [53] introduced a method combining sparse feature representation with dynamic slow feature analysis (SFA) and LSTM for fault detection and diagnosis in AHUs. This method resolves the issues of poor interpretability and low fault detection accuracy of traditional SFA when handling the time-varying dynamic characteristics of AHUs. Wang et al. [54] proposed a correlation-graph convolutional neural network (correlation-graph-CNN) method for fault diagnosis of wind turbines in building energy systems. This approach tackles the challenges of weak fault correlation and multidimensional parameters in wind turbine fault diagnosis, significantly improving diagnostic accuracy. Du et al. [55] presented a novel knowledge-embedded deep belief network (DBN) method, combining an extreme learning machine (ELM) and K-nearest neighbour (KNN) algorithms, to diagnose electronic–thermal and thermal–thermal multi-faults in building chillers. This method addresses the difficulty of collecting training datasets for multi-fault scenarios, which often results in low efficiency for machine learning models. All references in this section are listed in Table 2.

Table 2. Applications for different machine learning methods in equipment fault diagnosis.

Types	Methods	Advantages	Disadvantages	References
Supervised Learning	SVM	(1) Can solve high-dimensional problems; (2) Handles interactions between features; (3) Strong generalisation ability.	(1) Inefficient when there are many observations; (2) No universal solution for nonlinear problems.	[42,43]
	DT	(1) Handles nonlinear relationships; (2) No need for feature scaling; (3) Results are easy to visualise and interpret.	(1) Prone to overfitting; (2) Sensitive to noise in the data; (3) Unstable, small changes in data may lead to different tree structures.	[44]
	KNN	(1) Low cost for retraining; (2) Computation time and space scale linearly with the size of the training set; (3) Suitable for automatic classification in large sample domains.	(1) Limited interpretability of outputs; (2) High computational cost; (3) A lazy learning method.	[45]
Unsupervised Learning	AE	(1) Strong generalisation, no need for labeled data.	(1) For anomaly recognition, training data must consist of normal data.	[46,47]
	Clustering	(1) No need for a training set; simple and fast algorithms.	(1) Difficult to interpret clustering results; (2) Sensitive to initial conditions and parameters.	[48–51]
Deep Learning	CNN	(1) Excels at handling 2D data; (2) Improves training speed and generalisation performance.	(1) High computational complexity; (2) Difficulty extracting global information; (3) Limited regions for learned features.	[52,54]
	LSTM	(1) Solves long-term dependency issues; suitable for various types of sequential data; (2) Performs better in complex tasks.	(1) High computational cost and long training time; (2) Hard to tune parameters, prone to overfitting; (3) Poor interpretability, making debugging and optimisation difficult.	[53]
	DBN	(1) Layer-wise unsupervised pretraining; (2) Strong generalisation ability; (3) Avoids local optima.	(1) High training complexity; (2) Prone to overfitting; (3) Poor interpretability.	[55]

3. Machine Learning Applications in Building Energy Consumption Prediction

Building energy consumption prediction plays a crucial role in optimising energy management, reducing operational costs, supporting building design and planning, promoting sustainable development, improving building operation and management, and ensuring building safety and comfort. Traditional methods in this field often face limitations such as reliance on expert knowledge, the curse of dimensionality, and poor adaptability to new data.

Machine learning offers advantages such as automatic feature extraction, handling high-dimensional data, accurately capturing nonlinear relationships, and adapting to multi-time-scale predictions. This section reviews the research progress in predicting electrical load, cooling load, and heating load in building energy systems, as discussed in Sections 3.1–3.3, respectively.

3.1. Building Energy System Electrical Load Prediction

3.1.1. Traditional Machine Learning Techniques

To improve the accuracy and interpretability of electrical load predictions, many researchers have explored the application of various machine learning models. Peng, Lei,

Zhang, En, and Huang [8] developed a linear regression model to predict short-term building electricity load based on historical power consumption, weather, and date information. However, it struggled to capture nonlinear relationships, had limited generalisation ability, and heavily relied on data completeness. Zuazua-Ros et al. [56] used a random forest model to predict electricity load using multidimensional data, improving accuracy and reliability. Nevertheless, it required extensive data for training and had limited interpretability, and its performance was significantly influenced by feature selection and parameter tuning. Pradhan et al. [57] proposed a model based on support vector machines (SVMs) that analysed multiple factors to enhance prediction accuracy and generalisation ability. However, it was time-consuming to train, especially on large datasets, and its performance was highly dependent on parameter selection and kernel function determination. While these models offer distinct advantages, they also share some common challenges, such as requiring large amounts of training data; having high demands for data quality; limited generalisation ability; and being sensitive to factors like feature selection, parameter tuning, and network structure optimisation.

3.1.2. Deep Learning Techniques

To address the limitations of traditional machine learning techniques, many researchers have studied the application of deep learning in electrical load prediction. Pavlatos et al. [58] proposed a power load prediction method based on a bidirectional long short-term memory (BiLSTM) neural network, analysing its theoretical foundation and comparing its performance with architectures like recurrent neural networks (RNNs), long short-term memory (LSTM), and gated recurrent units (GRUs). The results showed that BiLSTM had significant advantages in predicting power load with a 24 h time step. Waheed et al. [59] applied an LSTM-based recurrent neural network (RNN) model for power load prediction, highlighting LSTM's ability to handle complex temporal relationships. This LSTM-RNN model demonstrated high accuracy and robustness in both hourly and annual load predictions. Arastehfar et al. [60] introduced a graph convolutional recurrent network (GCRN) model that improved the accuracy of short-term residential power load prediction. The model combined the spatial analysis capabilities of graph convolutional networks (GCNs) with the temporal sequence processing capabilities of RNNs. The experimental results indicated that its predictive performance was superior to that of traditional models such as convolutional neural networks (CNNs), especially when analysing complex building structure data. While these deep-learning-based electrical load prediction methods offer distinct advantages, they also share common challenges, including heavy reliance on data, high computational resource demands, sensitivity to parameter tuning and network structure selection, and limited generalisation ability due to training on specific building historical data. These factors impact their applicability to different buildings or climatic conditions.

3.1.3. Deep Learning Hybrid Models

In recent years, extensive research has been conducted on deep learning hybrid models. Liu et al. [61] proposed a hybrid prediction method combining time series generative adversarial networks (TimeGANs) with a convolutional neural network (CNN)-enhanced long short-term memory (LSTM) neural network. TimeGANs generate synthetic data to expand sparse datasets, the CNN optimises information extraction, and the data are input into the LSTM network for load prediction. Huang et al. [62] developed the CEEMDAN-TCN-ESN model for short-term power load prediction. It integrates complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN), temporal convolutional networks (TCNs), and echo state networks (ESNs), significantly improving prediction accuracy. Wu et al. [63] introduced a bidirectional gated recurrent unit–bidirectional long short-term mem-

ory (BiGRU-BiLSTM) hybrid model framework for power demand forecasting. It combines dual time-domain decomposition techniques, using CEEMDAN and K-means clustering to cluster and decompose data and variational mode decomposition (VMD) to process high-frequency sequences. Each component is predicted with the BiGRU-BiLSTM model, and the results are synthesised to achieve notable performance improvements. These hybrid models typically rely on large datasets and high computational resources. They also face challenges such as limited generalisation ability, high computational costs, insufficient network depth, and applicability restricted to specific time-scale load predictions.

Table 3 compares the application of different machine learning methods in load prediction. It highlights that traditional machine learning methods laid the foundation for early load prediction but revealed limitations as data volume and complexity increased. Deep learning methods have become mainstream, with hybrid models combining advantages to provide accurate predictions for complex scenarios.

Table 3. Comparison of methods for electrical load prediction.

Categories	Methods	Advantages	Disadvantages	References
Traditional Machine Learning	Linear Regression	(1) Simple model, strong interpretability; (2) Performs well on specific datasets.	(1) Cannot capture nonlinear relationships; (2) Limited generalisation ability.	[8]
	Random Forest	(1) Capable of handling complex datasets; (2) Captures nonlinear relationships.	(1) Requires large datasets for training and validation; (2) Less interpretable than linear models.	[56]
	SVM	(1) Handles high-dimensional data; (2) Can address nonlinear problems.	(1) Requires large datasets for training and validation; (2) Difficulties in selecting parameters and kernel functions.	[57]
Deep Learning	BiLSTM	(1) Strong predictive performance, excels at 24 h time-step predictions.	(1) Requires large datasets for training and validation; (2) Limited generalisation ability.	[58]
	LSTM-RNN	(1) High accuracy in both hourly and annual load predictions; (2) Robust even with incomplete or noisy data.	(1) Requires large datasets for training and validation; (2) Limited generalisation ability.	[59]
	GCRN	(1) Combines spatial analysis and time series processing; (2) Can analyze complex building structure data.	(1) Requires large datasets for training and validation; (2) Limited generalisation ability.	[60]
Deep Learning Hybrid Models	TimeGAN-TNN-LSTM	(1) TimeGAN generates synthetic data to expand sparse datasets; (2) CNN optimises information extraction.	(1) Relies on large datasets and high computational resources; (2) Limited generalisation ability.	[61]
	CEEMDAN-TCN-ESN	(1) TCN outperforms traditional models in time series feature extraction; (2) CEEMDAN enhances feature extraction.	(1) High computational cost; (2) Limited depth of TCN.	[62]
	BiGRU-BiLSTM	(1) CEEMDAN and K-means clustering decompose data; (2) VMD processes high-frequency sequences.	(1) Requires large datasets for training and validation; (2) Limited generalisation ability.	[63]

3.2. Cooling Load

3.2.1. Cooling Load Prediction Methods Based on Traditional Machine Learning

Wang et al. [64] proposed a hybrid prediction model combining a genetic-algorithm-optimised support vector machine (GA-SVM) and information granulation support vector machine (IG-SVM) to predict hourly cooling load and fluctuation range in large public

buildings. GA-SVM integrates the global search capability of GA with the predictive advantages of SVM, while IG-SVM leverages information granulation to enhance generalisation ability, effectively optimising SVM parameters to improve prediction accuracy. However, GA-SVM has long optimisation times and high costs for parameter tuning, IG methods are computationally complex and burdensome, and SVM has low training efficiency for large-scale datasets.

Zou et al. [65] proposed a random forest (RF)-based cooling load prediction model with a dynamic training set. The model uses RF to predict short-term cooling load from simulated data and then updates the training set with actual test data for retraining to predict the cooling load for subsequent days. This model demonstrates high prediction accuracy, strong robustness, and the ability to handle missing data effectively. The use of a dynamic training set improves real-time prediction capabilities. However, the model exhibits high computational complexity, large memory usage, longer prediction times, and a risk of overfitting.

Chen et al. [66] presented a physics-based multiple linear regression (LR) model for predicting hourly cooling load in buildings. This approach first establishes a physical heat transfer equation for cooling load and then converts it into a multiple linear regression model. The model has strong generalisation ability, short training time, and high interpretability. However, it also has high complexity, difficulties in feature selection, and stringent requirements for data quality.

3.2.2. Cooling Load Prediction Methods Based on Deep Learning

Deep learning models can automatically extract complex features from data, handle nonlinear and high-dimensional data, and enhance model generalisation and robustness. These models reduce overfitting through training on large datasets and regularisation techniques while improving resistance to outliers and noise through adversarial training.

Fan et al. [67] used a deep learning (DL) model to iteratively predict 24 h building cooling loads. By iteratively updating features, the model enhanced dynamic adaptability and effectively addressed short-term prediction issues. However, deep learning models require extensive data for training and are prone to overfitting with small datasets.

Song et al. [68] proposed a hybrid neural network model that integrates an attention mechanism (AM), bidirectional gated recurrent unit (BiGRU), convolutional neural network (CNN), and residual connection (RC) to predict short-term cooling loads in public buildings. The model outperformed common models but is complex, computationally expensive, time-consuming to train, and requires retraining and adjustment for different buildings and environments.

Han et al. [69] introduced a model combining multi-task learning (MTL) and bidirectional long short-term memory (BiLSTM) with a dynamic time warping (DTW) algorithm for short-term multivariable load prediction. The model performed well in cooling load predictions, but its accuracy was limited by incomplete utilisation of future load information. Additionally, the DTW algorithm faced efficiency and complexity challenges when processing large-scale data.

Morcillo-Jimenez et al. [70] presented a deep recurrent neural network (DRNN) comprising convolutional neural network (CNN) and long short-term memory (LSTM) layers to predict building energy consumption. The model handled complex time series data, with the CNN extracting features and LSTM modelling irregular trends, effectively predicting HVAC loads. However, it required substantial computational resources, limiting its applicability under resource constraints, and its decision-making process lacked transparency.

Table 4 compares the application of different machine learning methods in cooling load prediction. Traditional machine learning methods generally exhibit advantages in

adaptability and generalisation but have high computational complexity. In contrast, deep learning methods provide higher prediction accuracy and flexibility but still suffer from high computational complexity and strong dependence on data.

Table 4. Comparison of methods for cooling load prediction.

Type	Method	Advantages	Disadvantages	References
Traditional Machine Learning	SVM	(1) Strong parameter optimisation and generalisation ability; (2) High adaptability; (3) Capable of predicting hourly cooling loads and fluctuation ranges.	(1) Computationally complex; (2) High training cost; (3) Sensitive to missing data or outliers.	[64]
	RF	(1) High accuracy and robustness; (2) Strong ability to handle missing data; (3) Strong adaptability to training datasets.	(1) Computationally complex; (2) High memory usage; (3) Long prediction times; (4) Risk of overfitting.	[65]
	LR	(1) Strong generalisation ability; (2) Short training time; (3) High interpretability.	(1) High model complexity; (2) Difficult feature selection; (3) High data quality requirements.	[66]
Deep Learning	DL	(1) Strong ability to handle nonlinear and complex patterns; (2) High feature extraction capability and predictive performance; (3) High adaptability and flexibility.	(1) Complex computation, training, and parameter tuning; (2) High data requirements; (3) Risk of overfitting.	[67]
	AM, BiGRU, CNN, RC	(1) High prediction accuracy; (2) Reliable adaptive noise handling; (3) Reduced computational burden.	(1) High model complexity; (2) Strong data dependency; (3) Insufficient interpretability.	[68]
	MTL, BiLSTM	(1) High prediction accuracy; (2) Efficient data utilisation; (3) Strong ability to handle complex nonlinear relationships.	(1) Computationally complex and inefficient; (2) Task interference affects performance; (3) Prediction accuracy is limited by incomplete utilisation of future load information.	[69]
	DRNN	(1) Strong ability to handle complex time series data; (2) High feature extraction capability and predictive performance.	(1) High computational resource requirements; (2) Lack of interpretability; (3) Strong data dependency.	[70]

3.3. Heating Load

3.3.1. Heating Load Prediction Methods Based on Traditional Machine Learning

The XGBoost algorithm has been widely applied in various aspects of building heating load prediction. Wang et al. [71] proposed an XGBoost-based predictive control strategy for HVAC systems. XGBoost features strong interpretability due to its decision tree structure, fast runtime, and suitability for real-time predictions. However, it is sensitive to parameter tuning and data preprocessing, relying heavily on the quality and quantity of historical data. XGBoost is often combined with other machine learning techniques for heating load prediction. Cui et al. [72] introduced a hybrid prediction model combining XGBoost-RF for feature selection and CNN-GRU neural networks. XGBoost-RF selects predictive features, convolutional neural networks (CNNs) extract spatial features, and gated recurrent units (GRUs) capture temporal dependencies. The dual-channel structure integrates load variation trends and reduces cumulative errors. However, the model does not account for holiday effects or special events, limiting its comprehensiveness. Additionally, its interpretability and deployment efficiency require further optimisation. The support vector machine (SVM) is a supervised learning model that effectively handles nonlinear relationships and performs well with small datasets. Liu et al. [73] utilised an SVM learning model to predict and diagnose energy consumption in public buildings. The model demonstrated strong robustness and effective feature selection capabilities. However, it has

high computational complexity, parameter selection difficulties, and limited interpretability. The random forest (RF) model can learn the dynamic behaviour of buildings from historical data, avoiding the complex physical modelling process required in traditional model predictive control (MPC).

Smarra et al. [74] applied RF as the core of a data-driven predictive control (DPC) model, reducing dependence on complex physical models. RF demonstrated high prediction accuracy and strong scalability. However, it also exhibited high computational complexity, strong data dependency, and limited interpretability.

3.3.2. Heating Load Prediction Methods Based on Deep Learning

Building energy consumption data often involve multiple complex factors, such as weather changes and building usage patterns. Deep learning can handle these high-dimensional and nonlinear data, uncovering potential patterns and associations. Zhou et al. [75] proposed the ISSA-LSTM heating load prediction method, which is based on LSTM and optimised using the sparrow search algorithm (SSA). This approach accurately predicts heating loads; reduces energy consumption and costs; and addresses challenges like insufficient data, complex environments, computational complexity, and limited generalisation ability. However, it requires extensive parameter tuning and significant computational resources. Lv et al. [76] used LSTM as a baseline prediction model and compared it with bidirectional long short-term memory (Bi-LSTM) and gated recurrent units (GRUs). The study found that Bi-LSTM and GRUs outperform LSTM over long prediction horizons, effectively addressing time series characteristics and input parameter dependencies. However, the model still needs to identify suitable feature variables to improve prediction accuracy. Fouladfar et al. [77] applied artificial neural networks (ANNs) for adaptive heating load prediction in residential buildings, providing high-accuracy forecasts to optimise HVAC system operations and improve energy planning and operational management efficiency. However, the model is prone to overfitting and requires large amounts of labelled data for training, and its generalisation ability may decline under different building and environmental conditions. Quanwei et al. [78] introduced the MGMI deep learning model, combining multimodal graph (MG) attention networks and multi-scale informer (MI). The model uses an MG to capture spatial relationships and an MI to provide periodic trends and dependencies. Stationary wavelet transform (SWT) is employed for denoising. By integrating MG and MI structures, the model efficiently captures spatiotemporal information with high computational efficiency. However, it requires complex feature engineering and parameter tuning.

Table 5 compares the application of different machine learning methods in heating load prediction. Traditional machine learning methods generally exhibit high prediction accuracy and strong robustness but have drawbacks such as high computational complexity and poor interpretability. On the other hand, deep learning methods offer strong generalisation ability, high adaptability, and flexibility, but they still face challenges like high computational complexity and strong data dependency.

Table 5. Comparison of methods for heating load prediction.

Type	Method	Advantages	Disadvantages	References
Traditional Machine Learning	XGBoost	(1) High prediction performance; (2) Low computational cost; (3) Strong flexibility and adaptability; (4) Strong interpretability.	(1) High model complexity; (2) Strong data dependency; (3) Limited scalability.	[71]
	XGBoost-RF, CNN-GRU	(1) High prediction accuracy; (2) Low computational load; (3) Strong generalisation ability; (4) Effectively reduces cumulative errors.	(1) Limited feature selection; (2) High model complexity; (3) Poor interpretability.	[72]
	SVM	(1) Strong robustness; (2) High accuracy; (3) Effective feature selection capability.	(1) Computationally complex; (2) Difficult parameter selection; (3) Poor interpretability.	[73]
	RF	(1) High accuracy; (2) Strong robustness; (3) Strong scalability.	(1) Computationally complex; (2) Strong data dependency; (3) Poor interpretability.	[74]
Deep Learning	ISSA-LSTM	(1) Strong generalisation ability; (2) Low operational cost; (3) High prediction accuracy.	(1) High model complexity; (2) Strong data dependency.	[75]
	Bi-LSTM, GRU	(1) Captures bidirectional dependencies in time series; (2) Low prediction latency; (3) Strong adaptability.	(1) Computationally complex; (2) High implementation and maintenance difficulty; (3) Numerous model parameters.	[76]
	ANN	(1) High prediction accuracy and adaptability; (2) Strong generalisation ability; (3) Strong global search capability.	(1) Small datasets and strong data dependency; (2) Computationally complex; (3) Risk of overfitting; (4) High implementation and maintenance difficulty.	[77]
	MGMI	(1) Strong spatial relationship capture and multi-scale feature extraction; (2) Strong long-sequence time dependency capture; (3) High computational efficiency.	(1) High model complexity; (2) High computational resource consumption; (3) Poor interpretability.	[78]

4. Machine Learning Applications in Building Energy Planning and Operational Control

The application of machine learning in building energy planning and operational control is steadily advancing. By analysing historical building energy consumption data and occupant information, machine learning enables comprehensive monitoring and optimisation of building energy use. This chapter is divided into two sections. Section 4.1 reviews research on machine learning applications in building energy planning. Section 4.2 summarises the research status of machine learning in building energy operation and control.

4.1. Building Energy Planning

Building energy planning aims to design optimised multi-objective and multi-energy collaborative solutions to improve energy supply–demand matching efficiency and overall economic performance. However, with the increasing integration of renewable energy and the diversification of user load characteristics, building energy planning and operation face challenges such as multi-element, multi-dimensional, multi-objective, and multi-level coordination, as well as nonlinear system characteristics. During modelling and optimisation, it is essential to consider both continuous and discrete variables, time-dependent characteristics, and uncertainties in energy production, transmission, conversion, and consumption, significantly increasing system complexity and presenting greater planning challenges.

To address these issues, machine learning has become an important tool in building energy planning, primarily in three key areas: uncertainty simulation, scenario generation for planning, and optimisation of planning solutions.

4.1.1. Uncertainty Simulation

Uncertainty simulation focuses on modelling and analysing fluctuations in weather data, renewable energy output, and user demand. The goal is to provide a basis for system operation diversity and risk assessment. Traditional statistical methods, such as Markov chains and Monte Carlo methods, have commonly been used for uncertainty simulation. Fu [79] constructed probability density models for weather, photovoltaics, and HVAC systems using Markov chains and applied them to distribution network planning problems. Gabrielli et al. [80] introduced two-dimensional Monte Carlo simulation (2DMC) to separately handle inevitable randomness and reducible uncertainty. By incorporating correlation analysis of input parameters and percentile sensitivity analysis, this approach significantly improved the accuracy and reliability of risk assessment in building energy retrofit projects.

However, as energy system uncertainty increases in complexity, traditional statistical methods alone struggle to meet higher precision, robustness, and efficiency requirements. These methods often have limited modelling capabilities for high-dimensional nonlinear problems and low computational efficiency when handling multi-dimensional variable correlations.

To improve accuracy and efficiency in uncertainty modelling, researchers have introduced hybrid methods combining statistical models with machine learning. Zhu et al. [81] proposed a global sensitivity analysis method based on Monte Carlo simulation (MC) and tree Gaussian process (TGP) to quantify building load uncertainty. It identified key variables affecting building thermal performance using energy simulation models and observational studies. Tian et al. [82] adopted a Bayesian approach combined with Markov chain Monte Carlo (MCMC) techniques. Through sensitivity analysis and regression modelling, the study calibrated uncertain input parameters in building energy models, improving uncertainty quantification and predictive accuracy.

As the complexity of energy system uncertainty modelling continues to increase, researchers are shifting towards more efficient and accurate neural-network-based models to address high-dimensional nonlinear system challenges. Compared to traditional statistical methods and basic machine learning approaches, neural network models excel in capturing complex nonlinear relationships and spatiotemporal characteristics. Bampoulas et al. [83] applied a Bayesian convolutional neural network (BCNN) combined with a sliding window technique. By embedding Bayesian inference into a deep learning framework, the study effectively captured both stochastic and cognitive uncertainties related to energy conversion device operation and temperature deviations. This resulted in more precise, statistically interpretable real-time predictions. However, the study relied on synthetic data calibrated from physical models rather than real residential operational data, which may not fully reflect the complexity and randomness of real-world scenarios. Zhou et al. [84] addressed this limitation by introducing a hybrid uncertainty quantification method combining two-dimensional Markov chain Monte Carlo (2DMCMC) and a multi-layer neural network (MLNN) data-driven model. By integrating actual building energy system parameters with physical models, the study achieved efficient predictions of complex heat transfer and energy performance under multi-dimensional nonlinear input conditions. The method also enabled fast and reliable quantification of complex uncertainties, further improving the model's applicability and practicality. Table 6 presents a comparative analysis of the advantages and disadvantages of key machine learning methods used in building energy uncertainty simulation.

Table 6. Analysis of key machine learning algorithms for uncertainty simulation.

Methods	Characteristics	Advantages	Problem Solved	Limitations	References
TGP	Tree-structured hierarchical model based on Gaussian processes, capable of handling non-parametric regression and classification problems.	Accurately quantifies uncertainty; suitable for small datasets and global sensitivity analysis.	Used for uncertainty analysis of building loads, identifying key variables affecting building thermal performance.	Limited ability to handle high-dimensional data; high computational complexity.	[81]
MCMC	Combines Bayesian inference with MCMC sampling to estimate posterior probability distributions of parameters.	Improves uncertainty quantification accuracy; provides uncertainty ranges for parameters.	Calibrates uncertain input parameters in building energy models, enhancing model reliability through sensitivity analysis and regression modeling.	High computational complexity in parameter calibration; inefficient for large-scale data training.	[82]
BCNN	Integrates Bayesian inference with convolutional neural networks (CNNs), generating predictive distributions instead of point predictions; captures complex nonlinear spatiotemporal characteristics.	Quantifies prediction uncertainty; strong capability in modeling complex nonlinear relationships.	Real-time prediction of energy conversion device operations and temperature deviations, capturing stochastic and cognitive uncertainties.	Data are sourced from synthetic datasets calibrated using physical models, which may not fully reflect real-world scenarios.	[83]
MLNN	Deep learning model capable of capturing high-dimensional nonlinear data features and complex correlations.	Excellent modeling performance for high-dimensional nonlinear input–output relationships; capable of handling large-scale data.	Used in combination with 2D-MCMC for building energy system parameter prediction, enabling fast and reliable uncertainty quantification.	Requires large amounts of labeled data for training; high computational resource requirements.	[84]

4.1.2. Scenario Generation for Planning

In building energy planning, simulating and computing large-scale high-dimensional scenario data increase the complexity and difficulty of optimisation as decision variables and computational demands grow. However, most scenarios contain low-density valuable information, making it difficult for decision-makers to fully utilise them. By leveraging big data and machine learning techniques to extract typical operational scenarios, it is possible to provide efficient data support for planning and design, thereby enhancing the applicability and reliability of solutions.

Scenario generation methods typically rely on statistical techniques, with clustering techniques being a crucial method for scenario reduction. Clustering divides the entire scenario set into multiple groups and extracts representative features from each group, significantly reducing data volume while retaining key characteristics. Li et al. [85] applied the K-means clustering algorithm to generate representative energy demand scenarios for island residential communities and used scenario trees to capture multiple uncertainties in energy demand, providing more precise scenario data for energy system design and scheduling optimisation. Faraji et al. [86] compared K-means, K-medoids, and differential evolution algorithm (DEA) in day-ahead scheduling and operation. The study validated that under various conditions, K-medoids exhibited the best performance in preserving probabilistic information accuracy. Although hard clustering methods like K-means and K-medoids perform well in low-dimensional scenarios, their rigid classification approach limits adaptability to multidimensional dynamic systems. To improve model flexibility and adaptability, researchers introduced soft clustering methods that assign weights to scenarios based on their membership degree. Li et al. [87] proposed a fuzzy C-means clustering (FCM) method combined with an improved similar-day algorithm for scenario reduction. It also incorporated echo state networks (ESNs) and a multi-strategy Archimedes optimisation algorithm, significantly improving photovoltaic power forecasting accuracy in zero-energy buildings. Most clustering methods, however, use Euclidean distance as a similarity metric,

which fails to account for the shape characteristics of time series data. To address this issue, Liu et al. [88] introduced a hierarchical clustering and K-means-based scenario reduction method incorporating dynamic time warping (DTW) to measure time series similarity. The method selected representative operational days, effectively reducing the dataset size. Sayegh et al. [89] proposed a typical short-sequence generation-based scenario reduction method, combining K-medoids clustering and DTW, optimising representative time-period selection to significantly reduce data volume. Yang et al. [90] developed a custom clustering method for building energy consumption, using Pearson-correlation-based distance metrics and a novel centroid calculation method. This approach more effectively captured the shape characteristics of time series data, significantly improving scenario reduction while retaining temporal information.

Although clustering methods significantly improve scenario reduction, they heavily rely on large amounts of raw data and exhibit limitations in handling high-dimensional dynamic scenarios. To address these challenges, researchers introduced deep learning models that can learn data distributions directly to generate new scenarios. This approach reduces reliance on original data and enhances adaptability to high-dimensional nonlinear and complex dynamic scenarios.

Deep learning models learn distribution characteristics from historical data to generate high-quality new scenarios, addressing the shortcomings of scenario reduction methods that rely on raw data. The most widely applied deep learning techniques for scenario generation in power systems include variational autoencoders (VAEs) [91,92], long short-term memory (LSTM) [93,94], and generative adversarial networks (GANs) [95–97]. Qi, Hu, Dong, Fan, Dong, and Xiao [91] used VAE for scenario generation to solve power system planning and operation problems. By improving objective functions and evaluation metrics, the study mitigated common issues such as poor generalisation and overfitting in generative models. Gao, Zhi, Rong, and Yang [92] applied a VAE model to high-resolution power load data, performing dimensionality reduction and feature extraction. By integrating data from multiple sustainable residential buildings, the study compared the dimensionality reduction effectiveness of a VAE. However, the VAE struggled to generate high-quality and high-resolution dynamic scenarios. To enhance time series modelling, LSTM has been introduced. Wang, Liu, Zhang, Shen, Jing, and Xu [94] applied the LSTM algorithm to infer urban district energy performance. By learning the relationship between urban morphology parameters and energy performance, the model generated high-precision photovoltaic power generation and building total energy intensity data. GANs are widely used for high-dimensional complex scenario modelling. By employing adversarial training between generators and discriminators, GANs generate diverse and highly realistic scenario data. Tian, Li, Zhang, and Lv [96] used a GAN to generate parallel data from a small set of original building energy consumption sequences. The mixed dataset—combining original and generated artificial data—improved scenario generation performance. Reference [81] introduced a conditional GAN (CGAN) combined with CNN and Bi-LSTM for time series forecasting. By integrating historical data and random noise matrices into the generator, the model achieved high-precision photovoltaic power scenario generation. Mansour et al. [98] adopted a Wasserstein GAN (WGAN), optimising adversarial training stability using Wasserstein distance. The WGAN significantly improved scenario generation quality and diversity and was successfully applied to smart home energy management, optimising scheduling strategies and minimising household energy costs. Although GAN-based models generate high-quality scenarios, they require large-scale training data and high computational resources. To overcome these limitations, Silva et al. [99] proposed a nonlinear independent component estimation (NICE) model combined with convolutional layers as a flow-based generative model. NICE uses probabilistic flow transformations

to accurately capture true data distributions, generating highly realistic scenario samples. This method is particularly suitable for modelling complex building energy consumption scenarios and exhibited higher stability in high-precision load scenario generation.

NICE models provide reliable support for smart grids and building energy planning, making them a promising alternative to traditional GAN models.

4.1.3. Planning Scheme Optimisation

Planning scheme optimisation combines physical models and uncertainty scenarios to comprehensively characterise system features. This process includes model construction, uncertainty definition, constraint setting, and optimisation solving. Due to the nonlinear nature of multi-dimensional variables and complex constraints, traditional optimisation methods often face challenges in solution efficiency and achieving global optimality. Different studies have applied various optimisation algorithms to solve planning problems based on specific objectives. Zhan et al. [100] combined back propagation neural networks (BPNNs) with non-dominated sorting genetic algorithm II (NSGA-II) to optimise building retrofitting strategies. The objective was to minimise life-cycle carbon emissions, payback period, and indoor discomfort time. Sun et al. [101] applied Q-learning reinforcement learning to dynamically optimise energy scheduling strategies for the We-Energy system in an uncertain environment, demonstrating the advantages of reinforcement learning in adaptive optimisation for complex planning problems. Motamedi et al. [102] transformed transmission network planning into a four-layer optimisation problem and solved the transmission network expansion planning under power market conditions using Q-learning and search optimisation techniques.

While artificial intelligence techniques improve planning scheme efficiency, ensuring the interpretability of the generated plans is also crucial. Decision-makers must understand the solutions and potential risks to enhance transparency and robustness in decision making. Alova et al. [103] developed a gradient boosting decision tree (GBDT)-based machine learning framework to predict the success probability of power generation projects. The model used features such as plant capacity, fuel type, grid connection, ownership, and financing. The SHAP (Shapley additive explanations) method was applied to interpret results, enhancing model transparency. Sachit et al. [104] proposed an explainable artificial intelligence (XAI)-based suitability assessment method for optimising wind and solar power plant site selection. The study trained random forest, support vector machine (SVM), and multi-layer perceptron (MLP) models using data from over 55,000 real-world renewable energy plants. SHAP analysis was employed to explain the influence of technical, economic, environmental, and social factors on site selection decisions.

4.2. Operational Control

Reinforcement learning (RL) has been widely applied in building energy control in recent years due to its strong adaptability, high generalisation ability, and fast response time. RL methods are mainly categorised into model-free and model-based approaches.

In model-free methods, Touzani, Prakash, Wang, Agarwal, Pritoni, Kiran, Brown, and Granderson [9] applied the deep deterministic policy gradient (DDPG) algorithm for integrated control of HVAC and battery storage systems in real buildings. Compared to rule-based strategies, the DDPG control strategy reduced energy costs by 39.6% while maintaining thermal comfort. The advantage of the DDPG algorithm control strategy is its significant energy-saving effect, strong adaptability, and ability to directly interact with the environment without requiring an accurate mathematical model. However, this method has long training times and poor stability, and it requires a high level of technical expertise and computing resources. Gao, Matsunami, Miyata, and Akashi [12] applied the

DDPG algorithm for the continuous control of biomass generators, solar PV, and batteries in buildings. Compared to rule-based strategies, the DDPG algorithm exhibited the best performance under random and extreme conditions, while achieving the optimisation goals of off-grid operation of the energy system and safe operation of batteries. Sanaye and Sarrafi [105] applied the deep Q-network (DQN) algorithm to control the load of combined heat and power (CHP) units and the heat output of gas water heaters. Compared to two rule-based methods, this method reduced costs by 6.5% and 30.4%, while achieving an electricity self-sufficiency rate of 80.1%. Wan et al. [106] applied the DQN algorithm to control the amount of electricity purchased from the grid. Compared to an unoptimised scenario, the cumulative electricity cost was reduced by 11.38% using the DQN algorithm. Alabdullah and Abido [107] applied the double deep Q-network (double DQN) algorithm to control the active power output of power generation systems in buildings. This method achieved results close to the optimal solution while reducing computational time costs compared to mixed-integer linear programming (MILP) methods. The following studies primarily focus on model-based reinforcement learning methods. In model-based methods, Shen et al. [108] applied the double DQN algorithm to control the airflow ratio and battery state of charge in office building energy systems. This method enabled simultaneous control of various devices, reduced discomfort duration, and achieved a renewable energy utilisation rate of 82.1%. Dreher et al. [109] applied the proximal policy optimisation (PPO) algorithm to control the electrolyzer power of wind turbine–hydrogen storage systems. Among 70 cases with optimisation potential, PPO reduced opportunity costs in 43 cases. Abedi et al. [110] applied the Q-learning algorithm to control battery charging and discharging power in residential energy systems, effectively improving energy storage efficiency and reducing system operating costs. Claessens et al. [111] proposed a new method using convolutional neural networks (CNNs) to extract hidden state–time features, aiming to mitigate the curse of partial observability. In the supervised learning step of Q-iteration fitting, CNNs were used as function approximators to estimate the state–action value function (Q-function). This method was evaluated through qualitative simulation, which included a set of thermostatically controlled loads that shared only their air temperature, while their envelope temperature remained hidden. Simulation results showed that this method effectively captured hidden features and significantly reduced the electricity costs of the cluster.

5. Summary, Outlook, and Conclusions

Machine learning plays a critical role in improving fault diagnosis, load prediction, and energy planning/control in building energy systems. Different learning methods offer distinct advantages: supervised learning achieves high accuracy but depends on labelled data, unsupervised learning is effective for noisy and incomplete data but lacks precision, hybrid learning combines both strengths but is computationally intensive, and deep learning excels in handling time series data but requires large-scale datasets and high-performance computing. In load prediction, traditional machine learning models offer strong generalisation but face high computational complexity, whereas deep learning models improve prediction accuracy and adaptability but require extensive data and processing power. In energy planning and control, machine learning enhances the ability to manage system complexity, uncertainty, and dynamic changes, with reinforcement learning showing strong optimisation potential but facing challenges in interpretability and generalisation.

5.1. Summary

(1) In fault diagnosis, supervised learning methods achieve high accuracy when sufficient labelled data are available, but they rely heavily on large amounts of labelled data and have limited capability in capturing the dynamic characteristics of time series data. Unsupervised learning methods do not require labelled data and are suitable for handling noisy and incomplete data, but they have limitations in detection accuracy and adaptability to nonlinear data. Hybrid learning methods combine the advantages of supervised and unsupervised learning, making them suitable for complex scenarios, but the training process is complex and requires high computational resources. Deep learning performs well in processing time series data and modelling complex systems, but it strongly depends on large-scale labelled data and high-performance computing resources.

(2) In load prediction, traditional machine learning methods such as support vector machines (SVMs), random forest, and linear regression have strong adaptability and good generalisation ability but suffer from high computational complexity. Deep learning methods such as deep neural networks (DNNs) and long short-term memory (LSTM) show better prediction accuracy and flexibility but require large amounts of data and have high computational costs. In cooling load prediction, traditional machine learning methods like SVM, random forest, and linear regression have strong adaptability and generalisation ability but high computational complexity, whereas deep learning methods like DNNs and LSTM perform better in prediction accuracy and flexibility but require large amounts of data and high computing costs. In heating load prediction, traditional methods such as XGBoost and SVM have advantages in prediction accuracy and robustness but are sensitive to data quality and parameter tuning. Deep learning methods like LSTM and ANN effectively handle high-dimensional nonlinear data and have strong adaptability, but they also face challenges including data dependency and high computational resource requirements.

(3) In energy planning, machine learning significantly enhances the system's ability to handle complexity, dynamic changes, and uncertainty. Traditional building energy planning heavily relies on simplified mathematical models and static assumptions, which struggle to accurately capture the nonlinear interactions of multiple factors and environmental and usage condition changes. Machine learning, with its strong data-driven approach, can effectively address high-dimensional, nonlinear, and multi-objective optimisation problems, continuously improving planning accuracy and efficiency through intelligent modelling and historical data learning. In the field of energy control, reinforcement learning has strong optimisation capabilities, broad applicability, and fast response times, but it also faces challenges such as high data quality requirements, poor generalisation ability, and lack of interpretability.

5.2. Prospects

(1) In fault diagnosis, future trends will focus on multi-modal data fusion, integrating data from various sensors to enhance fault detection accuracy and robustness. Additionally, cross-disciplinary technology integration will drive fault diagnosis applications in various domains, enabling more efficient equipment management and maintenance.

(2) Increased automation and intelligence will develop self-learning and adaptive diagnostic systems for real-time monitoring and intelligent decision making. Big-data-driven fault diagnosis will use big data analysis to extract fault features and patterns, improving diagnostic accuracy and efficiency as data volume grows.

(3) In building energy planning, machine learning is expected to enable real-time analysis of large-scale data, identifying energy consumption patterns and predicting future

demand. This will lead to comprehensive monitoring and optimisation of building energy consumption, improving energy efficiency, reducing energy consumption, and lowering carbon emissions. It will also promote sustainable green building development, contributing to sustainability goals.

5.3. Conclusions

The research findings presented in this review have significant implications for the future of intelligent and sustainable building energy management. The integration of ML techniques in BES can lead to substantial improvements in fault detection, energy forecasting, and operational efficiency, contributing to reduced energy consumption and lower greenhouse gas emissions. The analysis highlights the transformative potential of deep learning and reinforcement learning in optimising energy scheduling and adaptive control strategies, enabling real-time decision making with high accuracy. Furthermore, the research underscores the importance of hybrid ML models and explainable AI techniques in enhancing model interpretability and reliability. These insights suggest that future research should focus on developing more robust, scalable, and adaptive ML frameworks to address the evolving complexities of BES. By leveraging ML-driven solutions, building energy management systems can achieve greater resilience, efficiency, and sustainability, aligning with global efforts toward carbon neutrality and smart infrastructure development.

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